

Vector Field Guided Path Following Control: Singularity Elimination and Global Convergence

WeiJia Yao, Héctor García de Marina and Ming Cao

Abstract—Vector field guided path following (VF-PF) algorithms are fundamental in robot navigation tasks, but may not deliver the desirable performance when robots encounter singular points where the vector field becomes zero. The existence of singular points prevents the global convergence of the vector field’s integral curves to the desired path. Moreover, VF-PF algorithms, as well as most of the existing path following algorithms, fail to enable following a self-intersected desired path. In this paper, we show that such failures are fundamentally related to the mathematical topology of the path, and that by “stretching” the desired path along a virtual dimension, one can remove the topological obstruction. Consequently, this paper proposes a new guiding vector field defined in a higher-dimensional space, in which self-intersected desired paths become free of self-intersections; more importantly, the new guiding vector field does *not* have any singular points, enabling the integral curves to converge globally to the “stretched” path. We further introduce the *extended dynamics* to retain this appealing global convergence property for the desired path in the original lower-dimensional space. Both simulations and experiments are conducted to verify the theory.

I. INTRODUCTION

The path following control problem is to find suitable control laws for a mobile vehicle to converge to and traverse along a prescribed geometric path, which is, mathematically, a one-dimensional manifold. Among many path following algorithms [1], vector field guided path following (VF-PF) algorithms are promising as they can achieve low path-following errors while requiring small control effort [1]. In VF-PF algorithms, a vector field is carefully designed such that its integral curves converge to and travel along the desired path [2]–[4]. Such a vector field is also known as a *guiding vector field* [5], since the desired velocity at each point of the field guides the robot.

A variety of VF-PF algorithms exist in the literature [2]–[5]. However, the existence of singular points where a vector field becomes zero compromises the global convergence to the desired path and also complicates the analysis of the algorithms [3]–[5]. Sometimes singular points are unavoidable. For example, if a 2D desired path is a simple closed curve, then at least one singular point always exists within the region enclosed by the desired path, as disclosed by the Poincaré-Bendixson theorem [6, Corollary 2.1]. The hairy

ball theorem also guarantees the existence of singular points of continuous vector fields defined on the sphere $\mathbb{S}^2$ [7, Theorem 13.32]. Therefore, for example, if the desired path is a circle, a robot cannot start and then escape from the center, which is a singular point of the 2D guiding vector field [4], [5]. In addition, the normalization of the vector field at this point is not well-defined. To the best of our knowledge, few studies deal with singular points of guiding vector fields. Some studies (e.g., [4]) assume that the singular points are repulsive to simplify the analysis. This assumption is dropped in [5] for the 2D case, but the extensibility of the integral curves might be finite if there are singular points. In general, path following algorithms only guarantee local convergence to the desired path (e.g., [8], [9]).

Following self-intersected desired paths is not achievable for existing VF-PF algorithms, since the crossing points of the desired path are also singular points of the vector field; hence, no guiding directions are available at the intersections. Thus, a robot gets stuck at the crossing points on the desired path (see Fig. 1). For this reason, the algorithms in [2]–[5] are ineffective simply due to the violation of the (implicit) assumption: no singular points are allowed to be on the desired path. In fact, many other path following algorithms are only applicable to simple desired paths, or have not addressed the problem of following self-intersected paths (e.g. [2]–[5]). For example, the classic line-of-sight (LOS) method [10]–[12] fails in this case as there is no unique projection point in the vicinity of a crossing point of the desired path.

When obstacle-populated environments are considered, a closely related line of research is feedback motion planning, where a feedback plan is equivalently a vector field defined on some configuration space [13, Chapter 8]. One of the most influential feedback motion planner is based on a navigation function [14]–[16], but the derived vector field always has undesirable singular points, due to the topology of the configuration space that is “punctured” by obstacles [14]. Therefore, eliminating these singular points in the *same* configuration space is infeasible.

Our contributions: As explained above, undesirable singular points of a vector field are common in many existing studies (e.g. [2]–[4], [14]). To tackle the above issues caused by singular points, we propose a new idea to change the topology of the desired path, and transform the guiding vector field into a singularity-free one in a *higher dimensional* configuration space. Consequently, we rigorously guarantee the global convergence of the vector field’s integral curves to the desired path, self-intersected or not. Therefore, we

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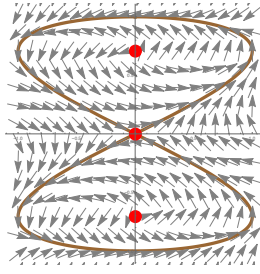


Fig. 1. The normalized vector field for a figure “8” path characterized by $\bar{\phi}(x, y) = x^2 - 4y^2(1 - y^2) = 0$. The three red points are singular points of the vector field in (2) where normalization is not well defined.

substantially improve the performance of the existing VF-PF algorithms. We also show that our proposed algorithm is a combined extension of both existing VF-PF and trajectory tracking algorithms. Note that our approach can be easily generalized for desired paths in a higher-dimensional space [17]. In addition, combining with the idea from [18], our approach can generate *singularity-free* vector fields for robot navigation in obstacle-populated environments, removing the undesirable singularity in [14].

Section II introduces the motivation of our work. In Section III, the 3D vector field and the problem definition are presented. The main results are elaborated in Section IV. In Section V, an experiment is described, and the comparison with a trajectory tracking algorithm is in Section VI. Section VII concludes the paper.

II. MOTIVATION

In the VF-PF problem, the desired path to follow is usually described by the zero-level set of a sufficiently smooth function. In particular, in the 2D Euclidean space $\mathbb{R}^2$, the desired path is described below [5]:

$$\bar{\mathcal{P}} = \{(x, y) \in \mathbb{R}^2 : \bar{\phi}(x, y) = 0\}, \quad (1)$$

where $\bar{\phi} : \mathbb{R}^2 \rightarrow \mathbb{R}$, called the *surface function*, is twice continuously differentiable to guarantee the existence and uniqueness [6, Theorem 3.1] of the integral curves of the guiding vector field introduced later. One can exploit the description (1) by using the absolute value of the surface function $|\bar{\phi}(p)|$ instead of the Euclidean distance $\text{dist}(p, \bar{\mathcal{P}}) := \inf\{\|p - q\| : q \in \bar{\mathcal{P}}\}$ between a robot's position $p \in \mathbb{R}^2$ and the desired path $\bar{\mathcal{P}}$ to measure how far the robot is away from the desired path. For example, when $|\bar{\phi}(p)| = 0$, the robot is precisely on the desired path. Thus, the 2D VF-PF problem is the design of a continuously differentiable vector field $\bar{\chi} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ to show up on the right-hand side of the differential equation $\dot{p}(t) = \bar{\chi}(p(t))$. The design requires to satisfy two conditions: (a) There exists a neighborhood $\mathcal{D} \subseteq \mathbb{R}^2$ of the desired path $\bar{\mathcal{P}}$ in (1) such that for all initial conditions $p(0) \in \mathcal{D}$, the distance $\text{dist}(p(t), \bar{\mathcal{P}})$ between the trajectory $p(t)$ and the desired path $\bar{\mathcal{P}}$ approaches zero as time $t \rightarrow \infty$; (b) If a trajectory starts from the desired path, then it stays on the path for $t \geq 0$.

The trajectories (solutions) of the differential equation $\dot{p}(t) = \bar{\chi}(p(t))$ are the *integral curves* of the vector field

$\bar{\chi}$. Therefore, we aim to find a suitable vector field $\bar{\chi}$ that satisfies the above two conditions. One example is [5]:

$$\bar{\chi}(x, y) = E \nabla \bar{\phi}(x, y) - k \psi(\bar{\phi}(x, y)) \nabla \bar{\phi}(x, y), \quad (2)$$

where $E = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, k is a positive constant, $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function with $\psi(0) = 0$, and $\nabla(\cdot)$ denotes the gradient of a scalar function $(\cdot)$. One of the assumptions is that there are no singular points of the vector field on the desired path. However, the vector field must have at least one singular point on a *self-intersected* desired path.

Theorem 1 (Crossing Points are Singular Points). *Suppose $c \in \bar{\mathcal{P}}$ is a crossing point of the desired path $\bar{\mathcal{P}}$ in (1). Then c is also a singular point of the vector field in (2); that is, $\bar{\chi}(c) = \mathbf{0}$.*

Proof. Since $c \in \bar{\mathcal{P}}$, we have $\bar{\phi}(c) = 0$, and thus $\bar{\chi}(c) = E \nabla \bar{\phi}(c)$ by (2). Next we show that the gradient at the crossing point $\nabla \bar{\phi}(c)$ is zero; hence $\bar{\chi}(c) = \mathbf{0}$. Suppose, on the contrary, the gradient is nonzero. Since $\nabla \bar{\phi}$ is continuously differentiable (as $\bar{\phi} \in C^2$), the implicit function theorem [19] concludes that there is a unique curve $\gamma : (-\delta, \delta) \rightarrow \mathcal{U}$ in a neighborhood $\mathcal{U}$ of c satisfying $\gamma(0) = c$ and $\bar{\phi} \circ \gamma(s) = 0$ for all $s \in (-\delta, \delta)$. But this contradicts the fact that $\bar{\mathcal{P}}$ is self-intersected. Therefore, the gradient at the crossing point is indeed zero. ■

In Fig. 1, the vector field at the crossing point is zero. We can attribute this to the loss of directional information at the crossing point: there is no obvious preference regarding which direction the trajectory should go towards. Thus, it is impossible to create a 2D vector field of which the integral curves follow the self-intersected desired path. In the following sections, we propose a solution to this inherent limitation by designing a new guiding vector field defined in the higher-dimensional Euclidean space $\mathbb{R}^3$.

III. PROBLEM FORMULATION

Starting from the description of the 2D (physical) desired path $\mathcal{P}^{\text{phy}}$ in (1), the (virtual) desired path $\mathcal{P}^{\text{hgh}}$ in $\mathbb{R}^3$ is naturally characterized by adding an additional constraint (or surface function) in $\mathbb{R}^3$ as follows:

$$\mathcal{P}^{\text{hgh}} = \{p \in \mathbb{R}^3 : \phi_1(p) = 0, \phi_2(p) = 0\}, \quad (3)$$

where $\phi_1, \phi_2 \in C^2$, $p = (x, y, w) \in \mathbb{R}^3$ and w is the additional dimension of the vector field. This implies that the desired path $\mathcal{P}^{\text{hgh}}$ is the intersection of two surfaces described by $\phi_i = 0, i = 1, 2$. The corresponding 3D guiding vector field $\chi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is [20]:

$$\chi(p) = \nabla \phi_1(p) \times \nabla \phi_2(p) - \sum_{i=1}^2 k_i \phi_i(p) \nabla \phi_i(p). \quad (4)$$

Here, the latter term $-\sum_{i=1}^2 k_i \phi_i \nabla \phi_i$ is the sum of signed gradients, providing a direction towards the two surfaces, and is intuitively called the *converging term*. The first term $\nabla \phi_1 \times \nabla \phi_2$, as the cross product of the two gradients, provides a tangential direction to each surfaces, and is intuitively called the *propagation term*. The set of all singular points of the

vector field, called the singular set, is formally defined as $\mathcal{C}^{\text{hgh}} = \{c \in \mathbb{R}^3 : \chi(c) = \mathbf{0}\}$. Some mild assumptions are necessary for the VF-PF problem [20]. We denote the converging term by $e_{\mathcal{M}}(\cdot) = -\sum_{i=1}^2 k_i \phi_i(\cdot) \nabla \phi_i(\cdot)$ and define the path-following error vector $e_{\mathcal{P}}(\cdot) = (\phi_1(\cdot), \phi_2(\cdot))$ in view of (3).

Assumption 1. There are no singular points on the desired path. More precisely, $\mathcal{C}^{\text{hgh}}$ is empty or otherwise there holds $\text{dist}(\mathcal{C}^{\text{hgh}}, \mathcal{P}^{\text{hgh}}) > 0$.

Assumption 2. As the norm of the path-following error $\|e_{\mathcal{P}}(p(t))\|$ approaches zero, the trajectory $p(t)$ approaches the desired path $\mathcal{P}^{\text{hgh}}$. Similarly, as the norm of the converging term $\|e_{\mathcal{M}}(p(t))\|$ approaches zero, the trajectory $p(t)$ approaches the union of the sets $\mathcal{P}^{\text{hgh}} \cup \mathcal{C}^{\text{hgh}}$.

See [20] for the precise mathematical expressions for Assumption 2. Note that Assumption 1 is violated for a self-intersected desired path by Proposition 1, but we will extend it in a higher-dimensional space such that it becomes non-self-intersected. These assumptions are made to avoid pathological consequences and ensure that $\mathcal{P}^{\text{hgh}}$ is a one-dimensional manifold [20], consistent with practical applications. Note that the nonlinear differential equation $\dot{p}(t) = \chi(p(t))$ is generally difficult to analyze. However, under the above assumptions, the integral curves of the 3D vector field (4) only have two outcomes [20]:

Lemma 1 (Dichotomy Convergence). *The integral curves of the guiding vector field χ in (4) converge to either the desired path $\mathcal{P}^{\text{hgh}}$, or the singular set $\mathcal{C}^{\text{hgh}}$.*

Since the ultimate objective is to follow the 2D physical desired path $\mathcal{P}^{\text{phy}}$, we need to project the higher-dimensional desired path $\mathcal{P}^{\text{hgh}}$ somehow. More specifically, a linear projection operator $P_a : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined; this operator is a linear map that can project a vector to the hyperplane orthogonal to a given nonzero vector $a \in \mathbb{R}^3$. With a slight abuse of notation, we use the same symbol for both this linear map and its matrix representation. Therefore,

$$P_a = I - \hat{a}\hat{a}^\top, \quad (5)$$

where I is the identity matrix of suitable dimensions and $\hat{a} := a/\|a\|$. The *projected desired path* is defined below:

$$\mathcal{P}^{\text{prj}} = \{q \in \mathbb{R}^3 : q = P_a \xi, \xi \in \mathcal{P}^{\text{hgh}}\}. \quad (6)$$

To let the integral curves of the vector field follow a self-intersected 2D desired path, there should be no singular points in the higher-dimensional guiding vector field (due to Proposition 1), which also implies the appealing feature of global convergence to the desired path (since $\mathcal{C}^{\text{hgh}} = \emptyset$ in Lemma 1). To sum up, the problem is formally formulated as follows:

Problem 1. Given a (possibly self-intersected) physical desired path $\mathcal{P}^{\text{phy}} \subseteq \mathbb{R}^2$, we aim to find a higher-dimensional desired path $\mathcal{P}^{\text{hgh}} \subseteq \mathbb{R}^3$, which satisfies the following

conditions¹:

- 1) There exist functions $\phi_1, \phi_2 \in C^2$ such that $\mathcal{P}^{\text{hgh}}$ is described by (3);
- 2) The singular set $\mathcal{C}^{\text{hgh}}$ of the corresponding higher-dimensional vector field $\chi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in (4) is empty.
- 3) There exists a projection operator P_a in (5) such that the projected desired path $\mathcal{P}^{\text{prj}}$ in (6) satisfies $\mathcal{P}^{\text{prj}} = \{(x, y, 0) \in \mathbb{R}^3 : (x, y) \in \mathcal{P}^{\text{phy}}\}$;

The essential approach is re-designing functions ϕ_i corresponding to the new desired path $\mathcal{P}^{\text{hgh}}$ such that the guiding vector field in (4) is singularity-free. In the next section, we propose a new idea to seek functions ϕ_i , and hence a higher-dimensional desired path $\mathcal{P}^{\text{hgh}} \subseteq \mathbb{R}^3$.

IV. SELF-INTERSECTED DESIRED PATH FOLLOWING

This section is split into three subsections regarding the solution to Problem 1. Firstly, we introduce the *extended dynamics* related to higher-dimensional vector fields. Secondly, we provide the detailed construction of the 3D (virtual) desired path $\mathcal{P}^{\text{hgh}}$ satisfying the conditions in Problem 1. Once such a 3D desired path is obtained, the 3D vector field will be automatically generated by (4). Thirdly, a control algorithm is designed for a unicycle robot to follow a self-intersected desired path using the 3D vector field.

A. Extended Dynamics

We introduce the extended dynamics to derive a result about the integral curves of the projected vector field $P_a \chi$.

Lemma 2 (Extended Dynamics). *Let $\chi : \mathcal{D} \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field that is locally Lipschitz continuous. Suppose $p(t)$ is the unique solution to the initial value problem $\dot{p}(t) = \chi(p(t))$, $p(0) = p_0 \in \mathcal{D}$. Then $(p(t), p^{\text{prj}}(t))$, where $p^{\text{prj}}(t) = P_a p(t)$ and P_a is the projection operator in (5) associated with a given nonzero vector $a \in \mathbb{R}^3$, is the unique solution to the following initial value problem:*

$$\begin{cases} \dot{p}(t) = \chi(p(t)) & p(0) = p_0 \\ \dot{p}^{\text{prj}}(t) = P_a \chi(p(t)) & p^{\text{prj}}(0) = P_a p_0, \end{cases} \quad (7)$$

Moreover, if the solution $p(t)$ converges to some set $\mathcal{A} \neq \emptyset \subseteq \mathbb{R}^3$, then the projected solution $p^{\text{prj}}(t)$ converges to the projected set $\mathcal{A}' = \{q \in \mathbb{R}^3 : q = P_a \xi, \xi \in \mathcal{A}\}$.

Proof. Since χ is locally Lipschitz and $\|P_a\| = 1$, where $\|\cdot\|$ is the induced matrix 2-norm, it follows that $P_a \chi$ is also locally Lipschitz continuous. Therefore, $(p(t), p^{\text{prj}}(t))$, where $p^{\text{prj}}(t) = P_a p(t)$, is the unique solution to (7). Fix t , then $\text{dist}(p^{\text{prj}}(t), \mathcal{A}') = \inf\{\|P_a(p(t) - q)\| : q \in \mathcal{A}\} \leq \inf\{\|P_a\| \|p(t) - q\| : q \in \mathcal{A}\} = \text{dist}(p(t), \mathcal{A})$. Since $p(t)$ converges to $\mathcal{A}$, $\text{dist}(p(t), \mathcal{A}) \rightarrow 0$ as $t \rightarrow \infty$. Namely, for any $\epsilon > 0$, there exists a $T > 0$, such that for all $t \geq T$, $\text{dist}(p(t), \mathcal{A}) < \epsilon$; hence $\text{dist}(p^{\text{prj}}(t), \mathcal{A}') \leq \text{dist}(p(t), \mathcal{A}) < \epsilon$. So $\text{dist}(p^{\text{prj}}(t), \mathcal{A}') \rightarrow 0$ as $t \rightarrow \infty$. Thus the projected solution $p^{\text{prj}}(t)$ converges to the projected set $\mathcal{A}'$. ■

¹Topologically, the desired path is one-dimensional, independent of the dimensions of the Euclidean space where it is. However, for convenience, a desired path is called n -dimensional if it lives in the n -dimensional Euclidean space $\mathbb{R}^n$ and not in any lower-dimensional subspace $\mathcal{W} \subseteq \mathbb{R}^n$.

B. Construction of the 3D Virtual Desired Path

Suppose a 2D (physical) desired path $\mathcal{P}^{\text{phy}}$ is parameterized by

$$x = f_1(w), \quad y = f_2(w), \quad (8)$$

where $w \in \mathbb{R}$ is the parameter of the path and $f_i \in C^2, i = 1, 2$. Then, we can simply let

$$\phi_1(x, y, w) = x - f_1(w), \quad \phi_2(x, y, w) = y - f_2(w), \quad (9)$$

such that the 3D virtual desired path is described by (3) and (9). Intuitively, the 3D desired path $\mathcal{P}^{\text{hgh}}$ is obtained by *stretching* the 2D desired path $\mathcal{P}^{\text{phy}}$ along the virtual w -axis (see Fig. 2(a)). Thus the first condition of Problem 1 is satisfied. One can calculate that $\nabla\phi_1 = (1, 0, -f_1'(w))^\top$ and $\nabla\phi_2 = (0, 1, -f_2'(w))^\top$, where $f_i'(w) := \frac{df_i(w)}{dw}, i = 1, 2$. Thus

$$\nabla\phi_1 \times \nabla\phi_2 = (f_1'(w), f_2'(w), 1)^\top.$$

Note that the third entry of this vector is a constant 1 regardless of the specific form of the desired path. This means that $\|\nabla\phi_1 \times \nabla\phi_2\| \neq 0$ in $\mathbb{R}^3$ globally. A closer examination of the guiding vector field (4) reveals that the propagation term is orthogonal to the converging term due to the property of the cross product. Therefore, since $\|\nabla\phi_1 \times \nabla\phi_2\| \neq 0$ in $\mathbb{R}^3$ globally, an appealing property is that the vector field $\chi(p) \neq 0$ globally in $\mathbb{R}^3$. This means that the guiding vector field χ has no singular points. Therefore, the second condition of Problem 1 is met. Furthermore, the 2D desired path $\mathcal{P}^{\text{phy}}$ is the projection of the 3D (virtual) desired path $\mathcal{P}^{\text{hgh}}$ on the plane $w = 0$. So the projection operator P_a can be chosen to associate with the vector $a = (0, 0, 1)^\top$. Thus, the third condition of Problem 1 is also satisfied. Now we can state the following theorem.

Theorem 2. Consider a 2D desired path $\mathcal{P}^{\text{phy}} \subseteq \mathbb{R}^2$ parametrized by (8). Let ϕ_1 and ϕ_2 be chosen as in (9). Then there are no singular points in the corresponding three-dimensional vector field $\chi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Let $a = (0, 0, 1)^\top$ for the projection operator P_a . Suppose the projected solution to (7) is $p^{\text{prj}}(t) := (x(t), y(t), w(t))^\top$. Then the 2D trajectory $p_2^{\text{prj}}(t) := (x(t), y(t))^\top$ will globally asymptotically converge to the 2D physical desired path $\mathcal{P}^{\text{phy}}$ as $t \rightarrow \infty$.

Proof. As discussed before, the singular set $\mathcal{C}^{\text{hgh}} = \emptyset$. According to Lemma 1 and Lemma 2, together with $\mathcal{C}^{\text{hgh}} = \emptyset$, the projected solution $p^{\text{prj}}(t)$ will globally asymptotically converge to the projected desired path $\mathcal{P}^{\text{prj}} = \{q \in \mathbb{R}^3 : q = P_a \xi, \xi \in \mathcal{P}^{\text{hgh}}\}$, where $\mathcal{P}^{\text{hgh}}$ is defined by (3) and (9). Since $a^\top p^{\text{prj}}(t) = a^\top P_a p(t) = 0$, the third coordinate of the projected solution $w(t) \equiv 0$, meaning that the trajectory $p^{\text{prj}}(t)$ lies on the XY -plane. Therefore, the 2D trajectory $p_2^{\text{prj}}(t) := (x(t), y(t))^\top$ will globally asymptotically converge to the 2D desired path $\mathcal{P}^{\text{phy}}$. ■

An example to explain Theorem 2 is shown in Fig. 2.

Remark 1. Three advantages of our approach are highlighted:

1) (Path Topology) By *stretching* along the virtual coordinate w , our approach transforms a possibly self-intersected

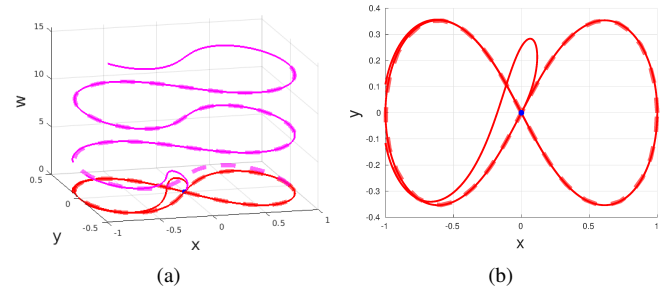


Fig. 2. A 2D self-intersected desired path $\mathcal{P}^{\text{phy}}$, of which the parametrization is $x = \cos w / (1 + \sin^2 w)$, $y = \sin w \cos w / (1 + \sin^2 w)$. (a) The magenta and red dashed lines are the higher-dimensional desired path $\mathcal{P}^{\text{hgh}}$ and the physical desired path $\mathcal{P}^{\text{phy}}$ respectively. The magenta solid line is the 3D trajectory $p(t)$ of $\dot{p} = \chi(p)$, and the red solid line is the projected trajectory $p_2^{\text{prj}}(t)$ in Theorem 2; (b) The projected trajectory $p_2^{\text{prj}}(t)$ in the XY -plane. The blue point is the starting point $(0, 0)$.

2D desired path $\mathcal{P}^{\text{phy}}$ to a non-self-intersected 3D virtual desired path $\mathcal{P}^{\text{hgh}}$, and thus Assumption 1;

2) (Singularity Elimination and Global Convergence) All singular points-whether they are saddle points or even stable nodes-in the original 2D guiding vector field $\bar{\chi}$ are eliminated in the 3D guiding vector field χ . Due to the singularity-free vector field χ , the global convergence to the 3D virtual desired path $\mathcal{P}^{\text{hgh}}$ is guaranteed, so it is for the 2D physical desired path $\mathcal{P}^{\text{phy}}$ by the extended dynamics.

3) (Surface Functions) Our approach facilitates the acquisition of the surface functions ϕ_i of which the intersection of the zero-level sets is the 3D desired path $\mathcal{P}^{\text{hgh}}$, once a parametric form of the 2D physical path $\mathcal{P}^{\text{phy}}$ is known. ◀

C. Control Algorithm Design for a Unicycle Robot

We can exploit the property of the global convergence of the guiding vector field's integral curves to the 2D desired path $\mathcal{P}^{\text{phy}}$. In particular, the control law design principle is to let a mobile robot's orientation eventually aligns with the direction indicated by the vector field. This principle implies that the guiding vector field only provides guidance signals rather than low-level control commands, and thus it is applicable to any robots whose motions are essentially determined by their orientations, such as the unicycle model (including the Dubin's car model), the car-like model and the underwater glider model [21]. The unicycle model is:

$$\dot{x} = v_u \cos \theta \quad \dot{y} = v_u \sin \theta \quad \dot{\theta} = \omega_u, \quad (10)$$

where (x, y) is the position, θ is the orientation, v_u is the speed control input and ω_u is the angular velocity control input. Since the 3D vector field will be used, the generalized 3D velocity vector of the robot needs to be defined as $\dot{p} = (\dot{x}, \dot{y}, \dot{w})^\top$, where $(\dot{x}, \dot{y})$ is the actual velocity of the robot as defined in (10) and $\dot{w}$ is the virtual velocity in the additional coordinate that is to be determined later. The control inputs v_u, ω_u and the virtual velocity $\dot{w}$ will be designed such that the orientation and the length of the generalized velocity $\dot{p} = (\dot{x}, \dot{y}, \dot{w})^\top$ will asymptotically become identical to the scaled vector field $s\hat{\chi}$, where s is a given positive constant.

Theorem 3. Suppose the model in (10) is considered, and a 2D parameterized desired path $\mathcal{P}^{\text{phy}} \subseteq \mathbb{R}^2$ is given. The

corresponding 3D vector field $\chi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is constructed in Section IV. Assume that the vector field satisfies $\chi_1(p)^2 + \chi_2(p)^2 \neq 0$ for $p \in \mathbb{R}^3$, where χ_i denotes the i -th entry of the vector field χ . Let

$$\dot{w} = s\hat{\chi}_3, \quad (11a)$$

$$v_u = s\|\chi^p\|, \quad (11b)$$

$$\omega_u = \dot{\theta}_d - k_\theta \hat{h}^\top E\hat{\chi}^p, \quad (11c)$$

$$\dot{\theta}_d = \left(\frac{-1}{\|\chi^p\|} \hat{\chi}^{p\top} E J(\chi^p) \dot{p} \right), \quad (11d)$$

where $(\hat{\cdot})$ is the normalization operator, s and k_θ are positive constants, $h = (\cos \theta, \sin \theta)^\top$, $\chi^p = (\hat{\chi}_1, \hat{\chi}_2)^\top$, $E = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $J(\chi^p)$ is the Jacobian matrix of χ^p with respect to the generalized position $p \in \mathbb{R}^3$ and $\dot{p} = (\dot{x}, \dot{y}, \dot{w})^\top$ is the generalized velocity. Denote the angle directed from $\hat{\chi}^p$ to $\hat{h}$ by $\beta \in (-\pi, \pi]$. If the initial angle $\beta(0) \in (-\pi, \pi)$, then the generalized velocity $\dot{p}$ will converge asymptotically to the scaled vector field $s\hat{\chi} = s(\hat{\chi}_1, \hat{\chi}_2, \hat{\chi}_3)^\top$ (i.e., $\beta(t) \rightarrow 0$). And the actual robot trajectory $(x(t), y(t))$ will converge to the physical desired path $\mathcal{P}^{\text{phy}} \subseteq \mathbb{R}^2$ asymptotically as $t \rightarrow \infty$.

Proof. We define two vectors in $\mathbb{R}^2$; i.e., $h' := v_u(\cos \theta, \sin \theta)^\top$ and $g' := s(\hat{\chi}_1, \hat{\chi}_2)^\top$. Also define the error (difference) between the generalized velocity $\dot{p}$ and the scaled vector field $s\hat{\chi}$ as below:

$$e_{ori}(t) = \dot{p} - s\hat{\chi} = \begin{bmatrix} v_u \cos \theta - s\hat{\chi}_1 \\ v_u \sin \theta - s\hat{\chi}_2 \\ 0 \end{bmatrix} = \begin{bmatrix} h' - g' \\ 0 \end{bmatrix} \in \mathbb{R}^3,$$

where the last entry 0 is due to (11a). Thus one only needs to focus on the first two entries of $e_{ori}(t)$. Note that $\|h'\| = s\sqrt{\hat{\chi}_1^2 + \hat{\chi}_2^2} = \|g'\|$, thus it is possible to proceed to show that $e_{ori}(t) \rightarrow 0$ asymptotically. In particular, it suffices to show that the orientation of h' asymptotically aligns with that of g' . Note that $\hat{h} = \hat{h}' = (\cos \theta, \sin \theta)^\top$ and $\hat{\chi}^p = \hat{g}' = (\hat{\chi}_1, \hat{\chi}_2)^\top / \sqrt{\hat{\chi}_1^2 + \hat{\chi}_2^2} = (\chi_1, \chi_2)^\top / \sqrt{\chi_1^2 + \chi_2^2}$. Let $e'_{ori} = \hat{h} - \hat{\chi}^p$. Choose the Lyapunov function candidate $V = 1/2 e'^{\top}_{ori} e'_{ori}$, and its time derivative is

$$\begin{aligned} \dot{V} &= \dot{e}'^{\top}_{ori} e'_{ori} = (\dot{\theta} E \hat{h} - \dot{\theta}_d E \hat{\chi}^p)^\top (\hat{h} - \hat{\chi}^p) \\ &= (\dot{\theta} - \dot{\theta}_d) \hat{h}^\top E \hat{\chi}^p \\ &\stackrel{(11c)}{=} -k_\theta (\hat{h}^\top E \hat{\chi}^p)^2 \leq 0. \end{aligned} \quad (12)$$

The second equation makes use of the identities: $\dot{\hat{h}} = \dot{\theta} E \hat{h}$ and $\dot{\hat{\chi}}^p = \dot{\theta}_d E \hat{\chi}^p$, where $\dot{\theta}_d$ is defined in (11d). The third equation is derived by exploiting the facts that $E^\top = -E$ and $a^\top E a = 0$ for any vector $a \in \mathbb{R}^2$. Note that $\dot{V} = 0$ if and only if the angle difference between $\hat{h}$ and $\hat{\chi}^p$ is $\beta = 0$ or $\beta = \pi$. Since it is assumed that the initial angle difference $\beta(t=0) \neq \pi$, we have $\dot{V}(t=0) < 0$, and there exists a sufficiently small $\epsilon > 0$ such that $V(t = \epsilon) < V(t=0)$. We will show by contradiction that $|\beta(t)|$ is monotonically decreasing with respect to time t . Suppose there exist $0 < t_1 < t_2$ such that $|\beta(t_1)| < |\beta(t_2)|$. It can be calculated that $V(t) = 1 - \cos \beta(t)$, and thus $V(t_1) <$

$V(t_2)$, contradicting the decreasing property of V . Thus $|\beta(t)|$ is indeed monotonically decreasing. By (12), $|\beta(t)|$ and $V(t)$ tends to 0, implying that the generalized velocity $\dot{p}$ will converge asymptotically to the scaled vector field $s\hat{\chi}$. Therefore, the generalized trajectory $(x(t), y(t), w(t))$ will converge to the higher-dimensional desired path $\mathcal{P}^{\text{hgh}}$. Then by Theorem 2, the actual robot trajectory $(x(t), y(t))$ (i.e., $p_2^{\text{pfj}}(t)$ in Theorem 2) will converge to the physical desired path $\mathcal{P}^{\text{phy}} \subseteq \mathbb{R}^2$ asymptotically as $t \rightarrow \infty$. ■

Remark 2. A significant feature of this algorithm is the introduction of a virtual coordinate and its dynamics. These do not correspond to any physical quantities in practice, but it is indispensable to form the extended dynamics in Lemma 2. By setting the third entry of the generalized velocity to be the same as the corresponding entry of the scaled vector field, it has been ensured that the third entry is always “aligned” and thus the algorithm mainly deals with the alignment of the other two entries with the scaled vector field. ◀

Remark 3. We explain why we do not consider constraints related to the control inputs or the desired path curvatures: As the robot is globally guided by the vector field, it can always re-orient its heading towards the desired path even if it temporarily deviates from the desired path due to these constraints. The fixed-wing aircraft experiment in [17] verifies the effectiveness of the control law. ◀

V. EXPERIMENT

An e-puck robot [22] is employed to follow a self-intersected desired path: the projection of a trefoil knot that is parameterized by $x = \cos(0.02w)(80 \cos(0.03w) + 160) + 600$, $y = \sin(0.02w)(80 \cos(0.03w) + 160) + 350$, where w is the parameter of the desired path. We use pixels as the distance unit. The robot has a data matrix on top, which is recognized by a overhead camera to obtain its position and orientation. The camera is connected to a computer, where the control algorithm runs. Then, the linear velocity and angular velocity control inputs v_u and ω_u are transmitted from the computer via a Bluetooth module at a fixed frequency of 20 Hz to the robot. The initial configuration of the robot is: $(x(0), y(0), \theta(0)) = (923, 545, \pi)$ and the initial value of the additional coordinate is $w(0) = 0$. The rest of the parameters are chosen to be $s = 10, k_1 = 0.5, k_2 = 0.2, k_e = 50$. The implementation procedure is listed in Algorithm 1, and the experimental results are shown in Fig. 3 and the supplementary material. At $t = 87s$, the robot was manually moved away from the desired path. After that, it headed towards the desired path to reduce the path-following error. As shown in Fig. 3(c), the norm of the path-following error characterized by $\|(\phi_1, \phi_2)\|$ decreased significantly and fluctuated around zero in the presence of sensor noise.

VI. DISCUSSION: PATH OR TRAJECTORY TRACKING?

In this section, we argue that our proposed approach is an extension of the trajectory tracking method, despite that the guiding vector field is the standard output for the path tracking (i.e., path following) approach. Therefore, our approach combines and extends elements from both algorithms.

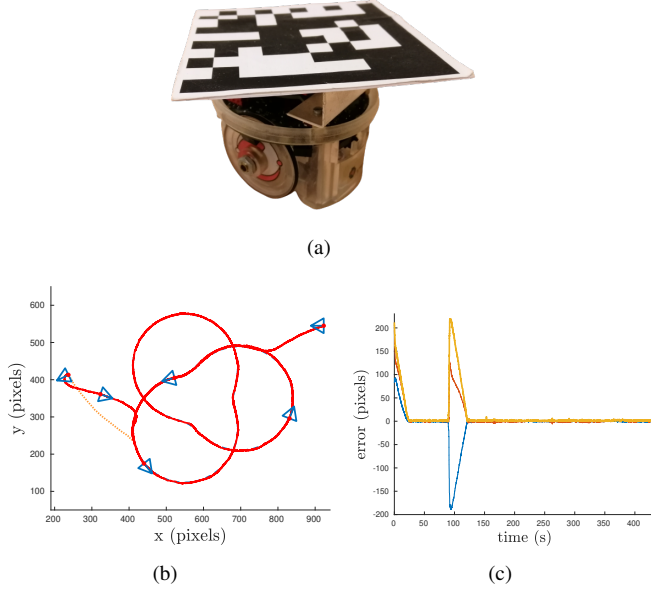
Algorithm 1 Proposed VF-PF control algorithm**Require:** Parametrization of the desired path in (8)Initialize the virtual coordinate $w(0) = w_0$.**while** stop signal not received **do**Obtain robot states (x, y, θ) and velocity $(\dot{x}, \dot{y})$.Calculate 3D vector field χ by (4) and (9).Calculate $\dot{w}$ by (11a) and let $\dot{p} = (\dot{x}, \dot{y}, \dot{w})^\top$.Calculate v_u and ω_u by (11b) and (11c).Apply the control inputs v_u and ω_u to the robot.Update $w \leftarrow w + \dot{w} \Delta t$, where Δt is the iteration lapses.**end while**

Fig. 3. Experiment results. (a) The e-puck robot; (b) Visualization of the experimental data. The red solid line is the actual robot trajectory, and the blue dashed line is the desired path (mostly covered by the robot trajectory). At $t = 87$ s the robot was manually moved to the leftmost position (indicated by the orange dashed line). The triangles represent the robot positions at different time instants, where the medians of the triangles pointing from the edge to the vertex indicate the headings; (c) The path-following errors ϕ_1 , ϕ_2 and $\|(\phi_1, \phi_2)\|$ represented by blue, red and yellow lines respectively.

The additional coordinate w in our proposed VF-PF algorithm seems similar to but in fact differs from the time t in trajectory tracking. In trajectory tracking, we remind that $\dot{t} = 1$; i.e., the desired trajectory $r(t)$ is prescribed, and it evolves as time elapses in an *open-loop* manner independent of the states of the robot. However, in our approach, the dynamics of the additional coordinate $\dot{w}(t) = s\hat{\chi}_3(p(t))$ in (11a) are in the closed-loop with the states of the robot (i.e., $p(t)$). Consequently, we improve the performance of our algorithm under noisy measurements, compared against the standard trajectory tracking approach. This claim is justified by numerical experiments and theoretical studies as follows.

Using the unicycle model in (10), we compare the proposed VF-PF algorithm in Section IV against the nonlinear trajectory tracking algorithm introduced in the classic monograph [21, p. 506]. For path following, we choose the desired path as the projection of a Lissajous knot parameterized as $x = 250 \cos(0.06w + 0.1) + 600$, $y = 250 \cos(0.08w + 0.7) + 350$, where $w \in \mathbb{R}$ is the parameter. Then we use (4),

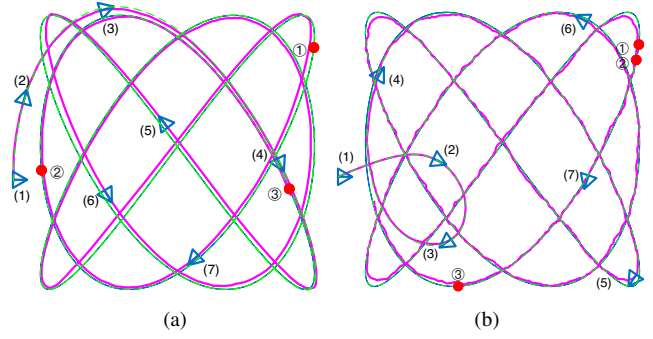


Fig. 4. (a) and (b) are simulation results with the proposed VF-PF algorithm and the nonlinear trajectory tracking algorithm respectively. The blue solid lines are the desired paths/trajectories; the green dash lines and magenta solid lines are the robot trajectories without white noise and with white noise added to the perceived robot positions respectively. The small blue triangles with labels represent a sequence of robot positions in the case with noise. The red points with labels are the *guiding points* for path following and *desired trajectory points* for trajectory tracking at $t = 0, 2, 30$ s respectively.

(9) to create the 3D vector field and (11) to calculate the control inputs v_u and ω_u . For trajectory tracking, the prescribed trajectory is obtained by replacing the path parameter w by time t . Recall that the dynamics of t are trivially the *open-loop* $\dot{t} = 1$, while the dynamics of the path parameter w are in the *closed-loop* (11a). The desired trajectory is denoted by $(x_d(t), y_d(t))$, and the feasible desired heading $\theta_d(t)$ is computed from $(x_d(t), y_d(t))$ [21, p. 503]. In both numerical simulations, the initial positions $(x(0), y(0))$ and orientations $\theta(0)$ of the robot are the same. For path following, the initial value of the additional coordinate is $w(0) = 0$, and for trajectory tracking, the initial time instant is $t = 0$. We also choose the same control gains for these two simulations.

If the measurements of the robot positions are accurate (i.e., no white noise is added), then both algorithms enable the robot to follow/track the desired path/trajectory accurately (see the green dash lines in Fig. 4). However, by checking the robot positions labeled by (1), (2) and (3) in Fig. 4(a) and Fig. 4(b), one notes that the robot's initial behaviors are quite different. In the trajectory tracking case, the robot revolves before aligning with the desired path, while in the VF-PF case, the robot moves more “naturally”: it “aligns” with the desired path immediately and reduces the distance to the desired path later. The “unnatural” movement for the trajectory tracking algorithm is due to the “open-loop” dynamics of the desired trajectory point. In the beginning, the desired trajectory point $(x_d(0), y_d(0))$ is far away from the robot's initial position (see the red point labeled by ① in Fig. 4(b)), and thus the robot needs to approach it to reduce the tracking error. Since the movement of the desired trajectory point is independent of the robot position, the robot needs to steer its heading continually as the trajectory point moves along the desired path; the desired trajectory points at $t = 2$ s and $t = 30$ s are illustrated as two red points labeled by ② and ③ respectively in Fig. 4(b). In view of (9), $(f_1(w(t)), f_2(w(t)))$ is termed the *guiding point* as the counterpart of the *desired trajectory point*. By contrast, although the guiding point $(f_1(w(t)), f_2(w(t)))$ is also on the far right side at $t = 0$ (the same as the desired

trajectory point), it “rushes” to the vicinity of the robot’s initial position shortly after 2 seconds (the red point labeled by ② in Fig. 4(a)), activated by the closed-loop dynamics $\dot{w}(t) = s\hat{\chi}_3(p(t))$ and the converging property of the integral curves of the vector field. This enables the robot to approach the “nearest point” of the desired path, avoiding the unnatural revolving behavior initially.

To compare the differences of the two algorithms under artificial sensor noise, we add a significant amount of band-limited white noise (power: 10, sampling time: 0.1 s) to the robot’s perceived positions. As seen from the magenta lines in Fig. 4, the robot trajectory for the path following algorithm is visually more smooth than that for the trajectory tracking algorithm. This is partly attributed to the propagation term in (4), which guarantees that there is always a nonzero tangential term to the level curves of the desired path, not being affected by the converging term due to orthogonality (see [23, Appendix] for the theoretical analysis). This property makes our algorithm appealing for fixed-wing aircraft motion control in the presence of sensor noise: By tuning the gains of the vector field such that the propagation term dominates, we can guarantee that the control will not demand the UAV’s heading to vibrate intensively, or even move backwards undesirably. By contrast, there is no tangential term in the trajectory tracking algorithm, and thus the robot’s heading vibrates persistently as the perceived position varies. In some extreme cases, the robot might move backward if the noisy perceived position overtook the desired trajectory point (see the supplementary material).

VII. CONCLUSION AND FUTURE WORK

We have proposed a 3D *singularity-free* guiding vector field to enable a robot to follow a self-intersected desired path in 2D by re-designing the surface functions. Moreover, we have rigorously proven the *global convergence* of the guiding vector field’s integral curves to self-intersected desired paths, which was further experimentally validated in path following tasks using a unicycle robot. Three advantages of our approach are summarized in Remark 1, regarding the path topology, singularity elimination and global convergence, and the surface functions. We have also compared our proposed VF-PF algorithm with a nonlinear trajectory tracking algorithm and the existing VF-PF algorithms, and concluded that our approach is a significant extension. Note that our proposed algorithm is applicable for 3D or even higher-dimensional desired paths (possibly in robot configuration spaces) [17]. Our idea can be utilized to create a singularity-free vector field for *global* robot navigation in obstacle-populated environments.

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